SAM Rover — End-User Product Guide & Quick-Start Manual

Welcome to your SAM Rover!
This guide will walk you through unboxing, powering on, connecting to Wi-Fi for the first time, and driving your rover from your phone, tablet, or laptop — no programming or technical setup required.

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1. What's in the Box

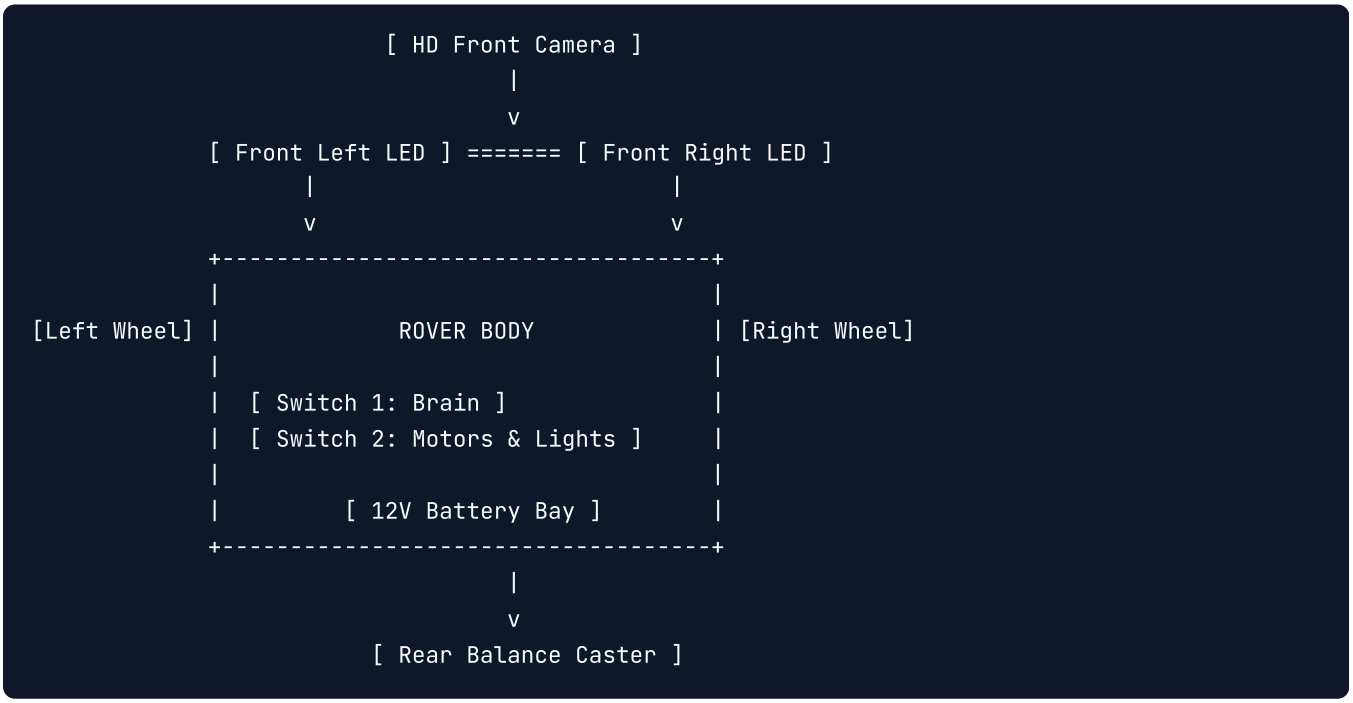
Before getting started, make sure your package contains the following items:

- **1x SAM Smart Mobile Rover** (fully assembled with HD camera, drive wheels, and front lights)
- **1× 12V Rechargeable Lithium-Ion Battery Pack**
- **1x Smart Battery Wall Charger**
- **1x Quick-Start Card** with your default access information

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2. Getting to Know Your Rover

Take a moment to familiarize yourself with the parts of your rover:



- **HD Camera**: Streams real-time video directly to your screen with ultra-low latency.
- **Front & Side LEDs**: Bright headlights and turn indicators for driving in low light.

- **Ultrasonic Eyes (Front & Sides)**: Smart distance sensors that automatically stop the rover before it hits an obstacle.
- **Switch 1 (Computer / Brain)**: Powers the onboard computer, Wi-Fi, and camera.
- **Switch 2 (Drive / Motors & Lights)**: Powers the motor drive wheels, headlights, and horn.
- **12V Battery Port**: Where your rechargeable battery plugs in securely.

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3. First-Time Power On & Setup

Your rover is powered by a high-capacity 12V Lithium-Ion battery with dual safety power switches. Follow these 3 simple steps to power it on:

```
[ Step 1: Connect Battery ]  ⟶  [ Step 2: Turn ON Switch 1 ]  ⟶  [ Step 3: Turn ON Switch 2 ]
  Plug 12V power jack              Computer starts (~25 sec)          Motors & lights ready!
```

Step 1: Connect the Battery

1. Place the rover on a flat, clear surface (such as a floor or table).
2. Insert the charged **12V battery pack** into the rover's battery bay.
3. Plug the battery power connector firmly into the rover's power socket.

Step 2: Turn ON Switch 1 (Rover Computer)

1. Flip **Switch 1** to the **ON** position.
2. A red light inside the rover will illuminate.
3. Wait about **25 to 30 seconds** for the rover's computer to boot up.
4. When ready, the rover will automatically broadcast its own private Wi-Fi network!

Step 3: Turn ON Switch 2 (Motors & Lights)

1. Flip **Switch 2** to the **ON** position.
2. The motor system and lights are now powered and ready to drive.

> **Tip:** Always turn **Switch 1** on first, and turn **Switch 2** off before storing your rover to conserve battery!

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4. Connecting to Your Rover's Wi-Fi

On first boot, your rover creates its own personal Wi-Fi network so you can connect directly without needing an internet router.

```
+-------------------------------------------------------------------------+
| FIRST-TIME WI-FI CONNECTION                                             |
|                                                                         |
| 1. Open Wi-Fi on Phone/Laptop ---> Select "SAM_ROVER_WIFI"              |
| 2. Enter Password: "raspberry"                                          |
| 3. Open Web Browser           ---> Go to: http://sam.local:8000         |
+-------------------------------------------------------------------------+
```

Step 1: Connect to the Rover's Wi-Fi

1. On your smartphone, iPad, tablet, or laptop, open your **Wi-Fi Settings**.
2. Look for the network named:
   `SAM_ROVER_WIFI`
3. Tap it and enter the default password:
   `raspberry`
4. Wait 3–5 seconds until your device displays the Wi-Fi connected checkmark. *(If your phone says "Internet not available", tap Keep Wi-Fi Connection — this is normal because the rover is communicating directly with your device).*

Step 2: Open the Control Dashboard

1. Open any web browser (Safari, Chrome, Firefox, or Edge).
2. Type the following address into the URL bar and press enter:
   `http://sam.local:8000`
   (If `sam.local` doesn't load on your device, type: `http://10.42.0.1:8000`)
3. The **SAM Rover Live Dashboard** will immediately open on your screen showing your live HD camera feed!

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5. Connecting Rover to Your Home or School Wi-Fi (Optional)

Once connected, you can easily link your rover to your home, office, or classroom Wi-Fi so that you don't have to switch Wi-Fi networks every time you want to drive!

```
[ Top Toolbar ]  ⟶  [ Tap Wi-Fi Icon ]  ⟶  [ Select Your Wi-Fi & Enter Password ]  ⟶  [ Con
```

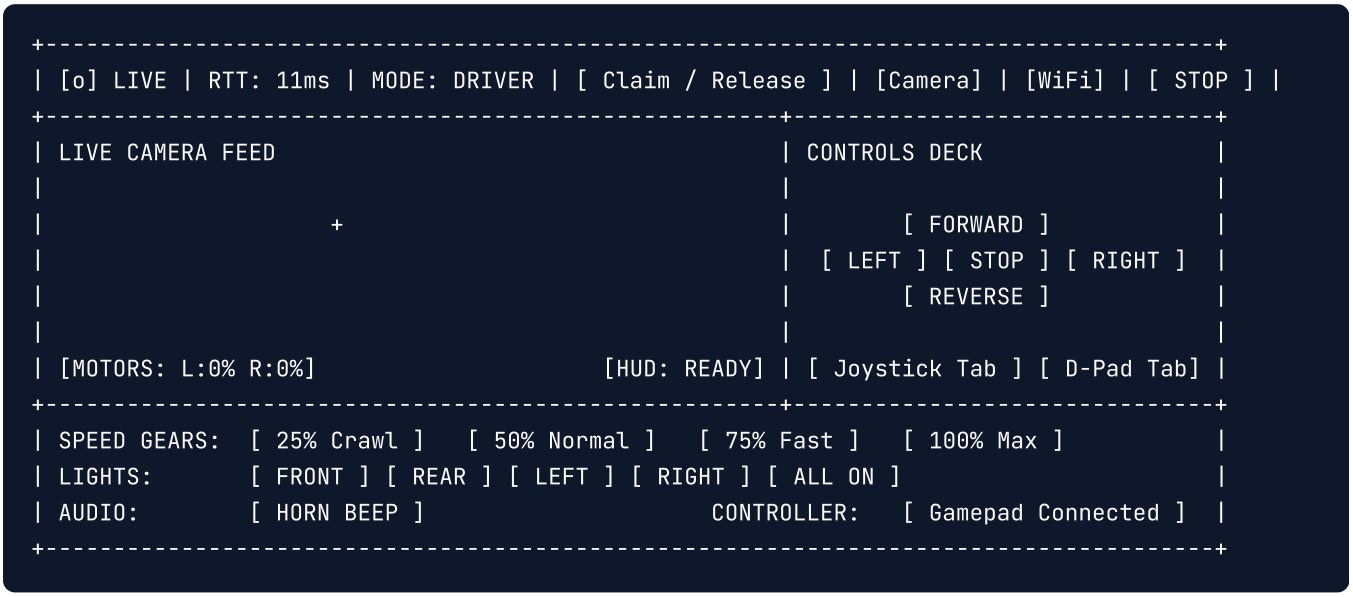
1. In the top navigation toolbar of the rover screen, tap the **Wi-Fi Icon** button (next to the camera snapshot icon).
2. The **Wi-Fi & Network Manager** window will open.
3. Tap `Scan Networks` to see all nearby Wi-Fi routers.
4. Select your home or school network from the list.
5. Type your Wi-Fi password and tap `Connect`.

6. Within a few seconds, the rover will connect to your router and display its new network address.
7. Now, any phone or laptop on your home Wi-Fi can open the rover by typing `http://sam.local:8000`!

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6. How to Drive & Control Your Rover

The dashboard is designed to be intuitive, responsive, and fun. Here is how to take control:



Step 1: Claim Control Authority

To prevent two people from driving at the same time:
* When you open the page, tap the amber `Claim` button in the top bar.
* The badge turns bright green and says `DRIVER (YOU)`.
* You now have complete driving control!

Step 2: Choose Your Favorite Control Method

You can drive using any of these 4 methods:

Method A: On-Screen Touch D-Pad (Phones & Tablets)

- Tap the big directional buttons on your screen:
- **Up Arrow**: Drive Forward
- **Down Arrow**: Drive Backward
- **Left Arrow**: Spin Left
- **Right Arrow**: Spin Right
- **Center Button**: Stop

Method B: Touch Virtual Joystick (Thumb Control)

- Tap the `Joystick` tab above the controls.
- Touch and drag the glowing cyan joystick knob with your thumb.
- Push forward to accelerate, pull back to reverse, and steer left/right smoothly with proportional turning.
- Release your thumb to stop instantly!

Method C: Keyboard (Laptops & Desktops)

- `W` or `Up Arrow`: Forward
- `S` or `Down Arrow`: Reverse
- `A` or `Left Arrow`: Spin Left
- `D` or `Right Arrow`: Spin Right
- `Spacebar`: Instant Emergency Brake
- `H`: Beep the Horn
- `L`: Toggle Headlights On/Off

Method D: Game Controller (Xbox, PlayStation, USB Gamepad)

Connect your favorite controller via Bluetooth or USB to your phone, tablet, or laptop:
* **Right Trigger (`RT`)**: Accelerate forward (gentle pull = slow, full pull = fast).
* **Left Trigger (`LT`)**: Brake / Reverse.
* **Left Thumbstick**: Steer left and right smoothly.
* `A` **Button** (Xbox) / `Cross` (PS): Beep Horn.
* `X` **Button** (Xbox) / `Square` (PS): Turn Headlights On/Off.
* `B` **Button** (Xbox) / `Circle` (PS): Emergency Stop.
* **Bumpers (`LB` / `RB`)** or **D-Pad Up/Down**: Shift speed gears up or down!

Step 3: Speed Gears (Choose Your Pace)

Tap any of the speed preset buttons to match where you are driving:

Gear	Speed	Best For
`25%` **(Crawl)**	Super Slow & Gentle	Driving through tight doorways, indoor precision, cautious parking.
`50%` **(Normal)**	Balanced Cruiser	Ideal default for living rooms, classrooms, and smooth floors.
`75%` **(Fast)**	Speedy Patrol	Hallways, driveways, and open spaces.
`100%` **(Turbo)**	Full 60 RPM Speed	Maximum hill climbing power, thick carpet, outdoor pavement.

Step 4: Lights & Horn

- **Headlights & Tail-lights**: Tap `Front`, `Rear`, `Left`, `Right`, or `All` to light up the path ahead.

- **Audio Horn**: Tap and hold the `Horn` button to sound the buzzer chirp — great for warning pets or people in the hallway!

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7. Built-in Smart Safety Features

Your SAM Rover is equipped with smart safety systems designed to protect the vehicle and your surroundings:

1. **Smart Automatic Obstacle Braking**:
2. If the rover detects a wall, table leg, or person in front closer than **35 cm**, it will automatically slow down.
3. If you get closer than **15 cm**, it will **automatically stop forward motion** to prevent a crash, while still allowing you to steer or reverse safely away!
4. **Connection Loss Safety (Watchdog)**:
5. If your phone battery dies, Wi-Fi drops, or you close the browser tab, the rover will **automatically stop moving within 1.5 seconds**. It will never run away out of control.
6. **Emergency Stop (STOP Button)**:
7. The bright red `STOP` button is always visible in the top-right corner. Tapping it halts all motors immediately.
8. **Auto-Release Handover**:
9. If you stop driving for 3 minutes, the rover automatically releases control so a friend or classmate can take a turn.

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8. Battery Charging & Daily Care

To keep your rover performing at its best, follow these battery care tips:

```
[ Battery Jack ] ---> [ Plug Into Included Wall Charger ] ---> [ Red = Charging | Green = Ready!
```

- **When to Recharge**: When the motors sound slightly weaker or the top dashboard bar shows a `LOW VOLT` badge, it is time to recharge.
- **Charging Time**: Disconnect the battery from the rover and plug it into the wall charger. The charger LED shines **Red** while charging and turns **Green** when full (typically 2 to 3 hours).
- **Storage**: When you are finished driving for the day, turn **Switch 2** OFF, turn **Switch 1** OFF, and unplug the battery connector. Store the rover in a dry, cool location.
- **Cleaning**: Wipe the wheels and camera lens with a clean microfiber cloth. Remove any carpet fibers or dust that collect around the wheel axles.

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9. Quick Troubleshooting Guide

What you see	Why it happens	Easy Fix
Cannot find `SAM_ROVER_WIFI` in Wi-Fi list	Rover is still booting up or battery is disconnected.	Check that the 12V battery is plugged in and **Switch 1** is turned ON. Wait a full 30 seconds for the Wi-Fi to appear.
Page says "Cannot open website" or times out	Device is not connected to the rover's Wi-Fi.	Open your phone's Wi-Fi settings. Confirm you are connected to `SAM_ROVER_WIFI`. Then go to `http://sam.local:8000` (or `http://10.42.0.1:8000`).
Camera shows video, but wheels don't turn when I press buttons	1. Control lease is not claimed. 2. Switch 2 is turned OFF.	1. Tap the amber `Claim` button in the top bar so it turns green. 2. Check that **Switch 2** (Motors & Lights) is flipped **ON**.
Rover moves very slowly or stops on carpet	Speed gear is set to 25% or battery is getting low.	Tap the `50%` or `100%` speed button. If still sluggish, plug the battery in to recharge.
"Observer Mode" message appears on screen	Another device is currently controlling the rover.	Have the other person tap `Release`, or tap `Take Control` to switch the driver to your device.
Rover drives slightly crooked when going straight	Normal friction differences between the left and right motor.	Slide the **Steering Trim** slider slightly left or right on the dashboard until it drives straight.
Front lights or horn don't work	Rover is in Standby mode.	Tap `Claim` to claim driver authority. The lights and horn will now toggle on demand.

10. Summary Quick Reference Card

Keep this summary handy for quick driving sessions:

```
1. Turn ON Switch 1 (Brain) & Switch 2 (Drive)
2. Connect phone/laptop Wi-Fi to: SAM_ROVER_WIFI (Password: raspberry)
3. Open browser and visit: http://sam.local:8000  (or 10.42.0.1:8000)
4. Tap the amber "Claim" button → Start driving!
```

Enjoy exploring with your SAM Rover!